# Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

### Division B | Meteorology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

**1. (2 points) Which satellite channel can show a jet stream in clear air?**

Unit: Observing severe weather: stations, balloons, radar, and satellites.
Worked-case data: A radar image shows a storm with reflectivity of 65 dBZ, a tight red-green velocity couplet on its southwest side, and a small area of CC values around 0.7 at the same spot. What is happening?
Separate investigation observations:
Reflectivity: A hook-shaped appendix on the storm's southwest side.
Velocity: Strong inbound and outbound winds side by side at the hook.
Correlation coefficient: A small area of 0.6-0.7 inside the hook.

A. Visible

B. Infrared only at night

C. Radar reflectivity

D. Water vapor

**Correct: D**

**2. (2 points) Unsaturated air rises and cools at about:**

Unit: Stability on Stüve and Skew-T diagrams.
Worked-case data: The surface temperature is 30 °C and the dew point is 22 °C. Estimate the LCL and comment on tornado potential.
Separate investigation observations:
Surface: Temperature 32 °C, dew point 23 °C.
Layer near 850 mb: A thin warm layer: temperature rises with height.
Above 700 mb: Temperature drops steeply with height.
Afternoon: A front lifts the air by midafternoon.

A. 9.8 °C per km

B. 2 °C per km

C. 6.5 °C per km always

D. 20 °C per km

**Correct: A**

### 3. (2 points) Which feature defines a supercell?

Unit: Thunderstorms and lightning.
Worked-case data: You see a flash and hear thunder 12 seconds later. How far away is the lightning, and are you safe outdoors?
Separate investigation observations:
Duration: The storm has lasted three hours.
Radar: A hook echo and a bounded weak-echo region.
Velocity: Persistent rotation in the mid levels.
Reports: Hail up to 2.75 inches (baseball size).

A. A warm front

B. A rotating updraft (mesocyclone)

C. A gust front

D. A single short-lived cell

**Correct: B**

### 4. (2 points) A hurricane with sustained winds of 115 mph is:

Unit: Hurricanes, typhoons, and cyclones.
Worked-case data: A hurricane with 140 mph sustained winds approaches a coast with a wide, shallow continental shelf at high tide. Rate the storm and assess surge.
Separate investigation observations:
Satellite: A clear eye surrounded by a ring of very cold cloud tops.
Recon aircraft: Sustained surface winds of 160 mph.
Motion: Slowing to 5 mph near the coast.
Coast: A wide, shallow bay ahead of the storm.

A. Category 4

B. Category 1

C. Category 3

D. Category 2

**Correct: C**

**5. (2 points) On Doppler velocity, green colors usually mean:**

Unit: Observing severe weather: stations, balloons, radar, and satellites.
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Separate investigation observations:
Reflectivity: A hook-shaped appendix on the storm's southwest side.
Velocity: Strong inbound and outbound winds side by side at the hook.
Correlation coefficient: A small area of 0.6-0.7 inside the hook.

A. Heavy rain

B. Hail

C. Motion toward the radar

D. Motion away from the radar

**Correct: C**

**6. (2 points) On a sounding, temperature and dew point lines close together indicate:**

Unit: Stability on Stüve and Skew-T diagrams.
Worked-case data: The surface temperature is 30 °C and the dew point is 22 °C. Estimate the LCL and comment on tornado potential.
Separate investigation observations:
Surface: Temperature 32 °C, dew point 23 °C.
Layer near 850 mb: A thin warm layer: temperature rises with height.
Above 700 mb: Temperature drops steeply with height.
Afternoon: A front lifts the air by midafternoon.

A. Very dry air

B. An inversion

C. Strong wind

D. Nearly saturated air

**Correct: D**

**7. (2 points) In the dissipating stage of a single-cell storm:**

Unit: Thunderstorms and lightning.
Worked-case data: You see a flash and hear thunder 12 seconds later. How far away is the lightning, and are you safe outdoors?
Separate investigation observations:
Duration: The storm has lasted three hours.
Radar: A hook echo and a bounded weak-echo region.
Velocity: Persistent rotation in the mid levels.
Reports: Hail up to 2.75 inches (baseball size).

A. The downdraft dominates

B. The updraft strengthens

C. The cloud is still growing

D. Rain has not begun

**Correct: A**

**8. (2 points) Where are a hurricane's strongest winds?**

Unit: Hurricanes, typhoons, and cyclones.
Worked-case data: A hurricane with 140 mph sustained winds approaches a coast with a wide, shallow continental shelf at high tide. Rate the storm and assess surge.
Separate investigation observations:
Satellite: A clear eye surrounded by a ring of very cold cloud tops.
Recon aircraft: Sustained surface winds of 160 mph.
Motion: Slowing to 5 mph near the coast.
Coast: A wide, shallow bay ahead of the storm.

A. Far ahead of the storm

B. In the eyewall

C. In the eye

D. In the outer rain bands

**Correct: B**

**9. (2 points) Why are cold, bright cloud tops on infrared images important?**

Unit: Observing severe weather: stations, balloons, radar, and satellites.
Worked-case data: A radar image shows a storm with reflectivity of 65 dBZ, a tight red-green velocity couplet on its southwest side, and a small area of CC values around 0.7 at the same spot. What is happening?
Separate investigation observations:
Reflectivity: A hook-shaped appendix on the storm's southwest side.
Velocity: Strong inbound and outbound winds side by side at the hook.
Correlation coefficient: A small area of 0.6-0.7 inside the hook.

A. They indicate clear skies

B. They mark tall thunderstorms

C. They mark fog

D. They show surface temperature

**Correct: B**

**10. (2 points) An environmental lapse rate between the moist and dry rates is:**

Unit: Stability on Stüve and Skew-T diagrams.
Worked-case data: The surface temperature is 30 °C and the dew point is 22 °C. Estimate the LCL and comment on tornado potential.
Separate investigation observations:
Surface: Temperature 32 °C, dew point 23 °C.
Layer near 850 mb: A thin warm layer: temperature rises with height.
Above 700 mb: Temperature drops steeply with height.
Afternoon: A front lifts the air by midafternoon.

A. Absolutely unstable

B. Always neutral

C. Conditionally unstable

D. Absolutely stable

**Correct: C**

**11. (2 points) Interpret the adjacent opposite radial velocities cautiously.**

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

answer: A tight adjacent inbound/outbound couplet is evidence consistent with rotation; geometry, resolution and quality control still matter. (2 points).

**12. (2 points) What additional interpretation is supported by low correlation coefficient collocated with the velocity couplet?**

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

answer: Mixed or irregular scatterers consistent with lofted debris can support a tornado-debris interpretation when collocated with rotation; use corroborating evidence. (2 points).

**13. (2 points) Calculate the surface temperature-dew-point spread.**

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

answer: 5 C (2 points).

Numerical target 5; absolute tolerance 0.01. Accepted units: , C.

**14. (2 points) Identify one reason that 65 dBZ reflectivity alone does not prove a tornado.**

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

answer: High reflectivity can indicate large or numerous precipitation targets, including hail; it does not directly establish a rotating ground-contact vortex. (2 points).

**15. (2 points) A storm moves 25 km east in 30 minutes. Find mean speed.**

Storm-track table:
Time | Eastward distance
0 min | 0 km
30 min | 25 km

answer: 50 km/h (2 points).

Numerical target 50; absolute tolerance 0.1. Accepted units: , km/h.

**16. (2 points) A rain gauge records 35 mm in 0.5 hours. Calculate mean rain rate.**

Gauge table:
Elapsed time | Accumulated rain
0 h | 0 mm
0.5 h | 35 mm

answer: 70 mm/h (2 points).

Numerical target 70; absolute tolerance 0.14. Accepted units: , mm/h.

**17. (2 points) A satellite IR time series shows cloud-top brightness temperature falling from 240 K to 220 K. Give a physically plausible interpretation and a limitation.**

IR table:
Frame 1 | 240 K
Frame 2 | 220 K

answer: Colder cloud tops can be consistent with cloud growth to higher levels, but retrieval, cloud type and environmental temperature profile affect interpretation. (2 points).

**18. (2 points) Station wind shifts form a coherent boundary while pressure rises behind it. Describe a candidate thunderstorm-related mechanism.**

Station table:
Ahead of boundary | Original wind, lower pressure
Behind boundary | Shifted wind, rising pressure

answer: A thunderstorm cold pool/outflow boundary is consistent with the observations; compare radar, temperature changes and timing rather than identifying it from one variable alone. (2 points).

**19. (2 points) Radar detects strong winds perpendicular to its beam but a small radial velocity. Explain why the wind need not be weak.**

Radar geometry table:
Wind direction | Perpendicular to beam
Radial velocity | Small
Independent wind observation | Strong

answer: Doppler radial velocity is only the component along the beam; perpendicular motion contributes little to that component. (2 points).

**20. (2 points) A short intense rain period falls on an already saturated basin. Explain why rainfall amount alone does not establish flood severity.**

Basin table:
Antecedent soil | Saturated
Rain | Short, intense burst
Catchment drainage/routing | Not supplied

answer: Antecedent moisture, infiltration, drainage, catchment area and routing affect runoff and flood response; evaluate rainfall intensity and basin conditions together. (2 points).

**21. (2 points) Use Figure F1. Read the velocity difference between the adjacent west and east cells.**

## Same small storm area: collocated measurements

| Region | Reflectivity | Radial velocity | Correlation |
|---|---|---|---|
| West cell | 65 dBZ | -30 m/s | 0.70 |
| East cell | 60 dBZ | +35 m/s | 0.72 |
| Background | 25 dBZ | +5 m/s | 0.98 |

F1. Synthetic dual-polarization radar panels (original synthetic teaching figure)

answer: 65 m/s (2 points).

Numerical target 65; absolute tolerance 0.195. Accepted units: , m/s.

**22. (2 points) Use Figure F1. Explain why the juxtaposed velocity signs support investigating rotation rather than concluding that both cells are calm.**

Refer to F1. Synthetic dual-polarization radar panels (original synthetic teaching figure) shown with the first question in this figure group.

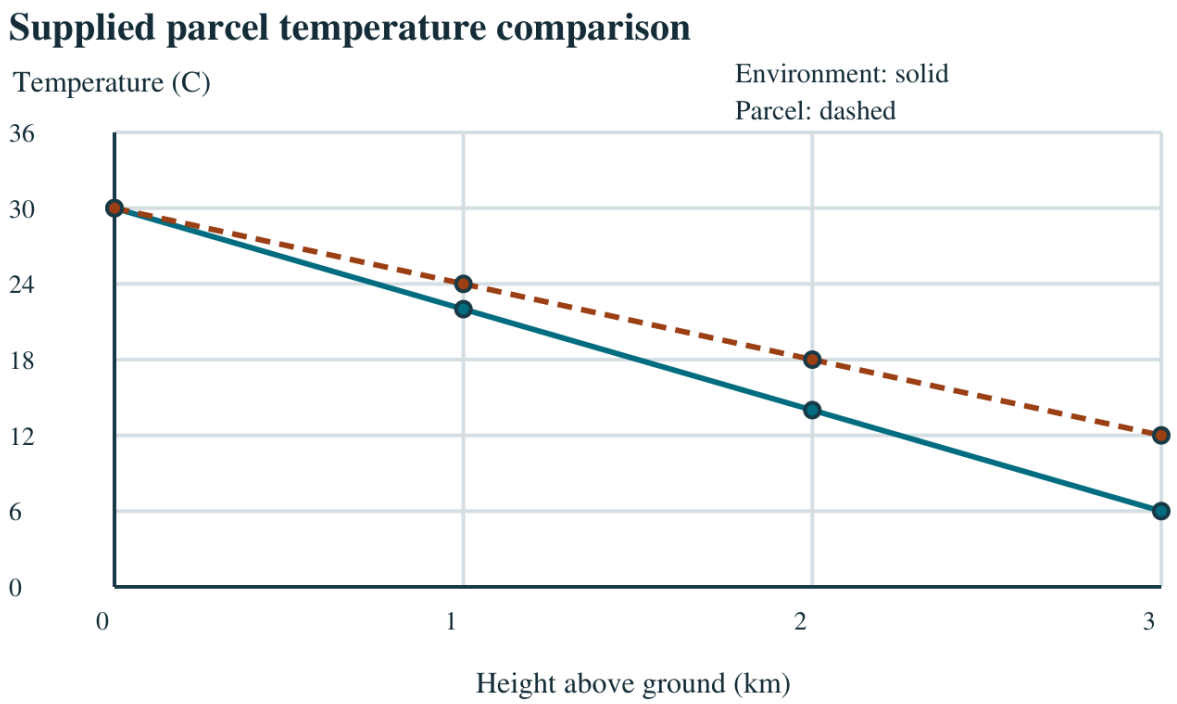
answer: The adjacent approaching and receding velocities form a couplet consistent with rotation; geometry, resolution and other evidence are needed for confirmation. (2 points).

**23. (2 points) Use Figure F1. What extra interpretation does low correlation in the same area support, and what does it not prove by itself?**

Refer to F1. Synthetic dual-polarization radar panels (original synthetic teaching figure) shown with the first question in this figure group.

answer: It can support mixed/irregular scatterers such as lofted debris when collocated with rotation, but cannot independently prove a ground-contact tornado without corroborating evidence. (2 points).

**24. (2 points) Use Figure F2. Read parcel-minus-environment temperature at 2 km.**




F2. Parcel and environment profiles (original synthetic teaching figure)

answer: 4 C (2 points).

Numerical target 4; absolute tolerance 0.012. Accepted units: , C.

**25. (2 points) Use Figure F2. Under the stated simplification, is the parcel positively or negatively buoyant at 2 km? Explain.**

Refer to F2. Parcel and environment profiles (original synthetic teaching figure) shown with the first question in this figure group.

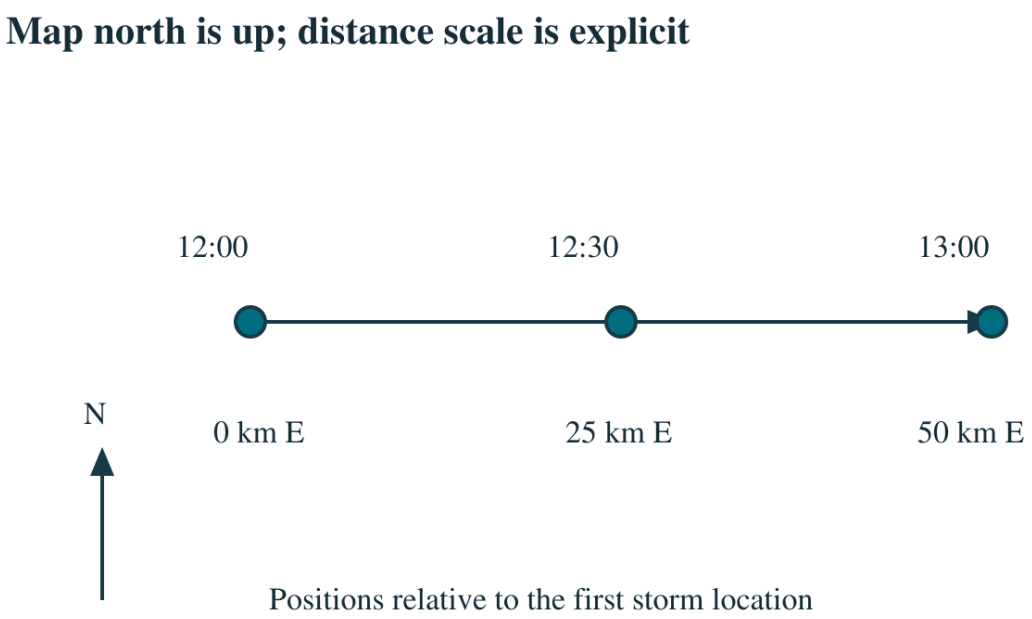
answer: Positively buoyant: the parcel is warmer and hence less dense than its environment at matched pressure under the stated simplification. (2 points).

**26. (2 points) Use Figure F2. Why cannot this profile alone establish tornado occurrence?**

Refer to F2. Parcel and environment profiles (original synthetic teaching figure) shown with the first question in this figure group.

answer: It does not supply all required storm and wind information, including shear/rotation, initiation and ground-level evidence; instability alone does not establish a tornado. (2 points).

**27. (2 points) Use Figure F3. Calculate mean storm speed from 12:00 to 13:00.**




F3. Storm-track map (original synthetic teaching figure)

answer: 50 km/h (2 points).

Numerical target 50; absolute tolerance 0.15. Accepted units: , km/h.

**28. (2 points) Use Figure F3. Give the direction of the plotted motion.**

Refer to F3. Storm-track map (original synthetic teaching figure) shown with the first question in this figure group.

answer: East. (2 points).

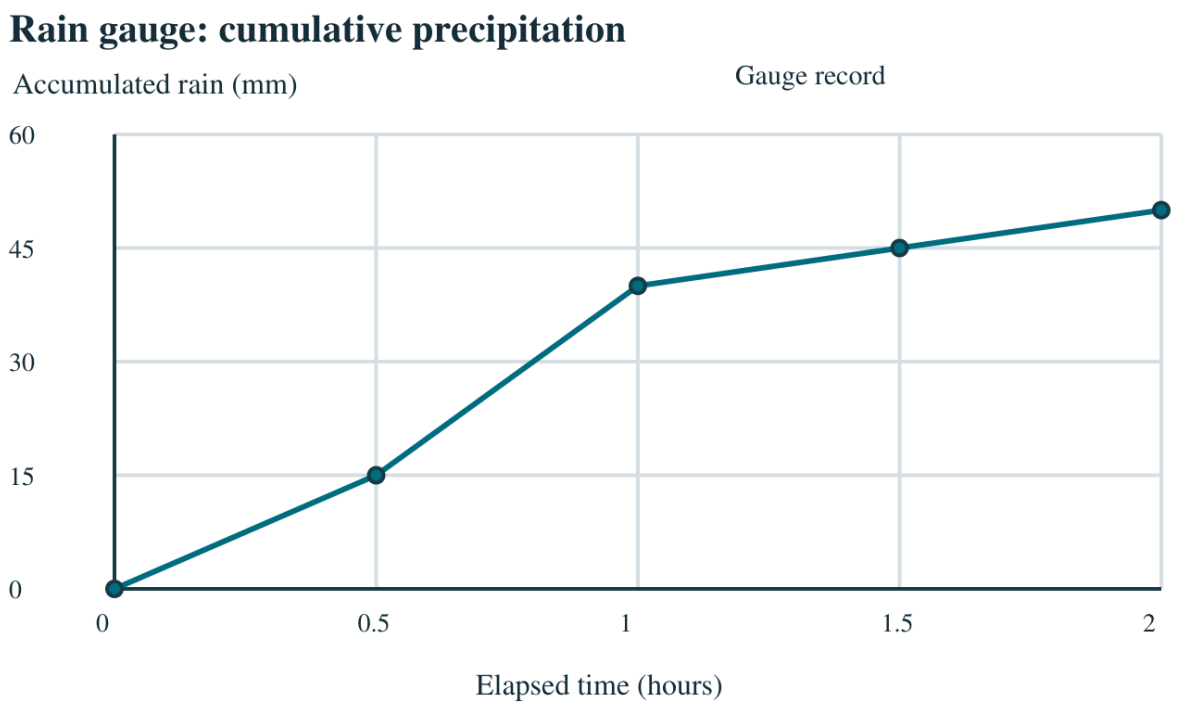
**29. (2 points) Use Figure F3. If the motion continues unchanged, how many minutes after 13:00 would a point 25 km farther east be reached?**

Refer to F3. Storm-track map (original synthetic teaching figure) shown with the first question in this figure group.

answer: 30 minutes (2 points).

Numerical target 30; absolute tolerance 0.09. Accepted units: , minutes.

**30. (2 points) Use Figure F4. Calculate the mean rain rate during the 0.5-to-1.0 h interval.**




F4. Accumulated rainfall curve (original synthetic teaching figure)

answer: 50 mm/h (2 points).

Numerical target 50; absolute tolerance 0.15. Accepted units: , mm/h.

**31. (2 points) Use Figure F4. Identify the interval with the steepest accumulation and explain what the slope measures.**

Refer to F4. Accumulated rainfall curve (original synthetic teaching figure) shown with the first question in this figure group.

answer: 0.5 to1.0h; slope is rainfall rate, and this interval has the largest increase per unit time. (2 points).

**32. (2 points) Use Figure F4. For an already saturated basin, explain why the steep interval deserves attention even if a later interval has a higher cumulative total.**

Refer to F4. Accumulated rainfall curve (original synthetic teaching figure) shown with the first question in this figure group.

answer: Short intense rainfall can create rapid runoff when infiltration capacity/storage is limited. Cumulative total and current intensity describe different aspects of flood response. (2 points).